

Appendix A: Proposed Scope of Work

Getting Fields in Shape for Fields of the World Phase 2 Proposed Scope of Work Clark University

The following is a proposal that outlines Clark's contribution to the next phase of FTW along with requested budget and anticipated timelines. The focal areas in this proposal are based on the assumption that PRUE and a semantic segmentation approach will remain the primary means for field boundary mapping within the FTW project. The proposal is based on discussions beginning on 5/1/2026, subsequent feedback on the draft, and points raised in the initial FTW v2 [document](#) developed by Hannah Kerner.

Focus Areas

Focus 1 - New labels

FTW's label catalog will be expanded, and there is a particular need to add labels from unsampled countries and those with harder-to-map agricultural systems, where there are no pre-existing field boundary data available. These labels will be collected as part of a field boundary digitizing campaign led by ASU that is separate from a planned Clark-led, NASA-funded effort that will focus on Africa using VHR imagery. Although Clark will contract ASU to collect labels for the NASA project, the team members and methods will be primarily separate from the FTW v2 label collection effort. Clark's role in the FTW v2 labeling campaign will therefore primarily be advisory, and will include some or all of the following aspects:

- Guidance on sample design and collection methodology and placing the sample;
- Assistance with labeling team training, including potential labeling tutorials;
- Helping to implement a labeling platform (assuming use of Clark's updated platform), or integrate key components (e.g. quality control; shape metrics) into ASU's existing labeling tools;
- Review collected labels as needed.

Focus 2 - Getting fields in shape

A key area of improvement for the FTW project is to improve the quality of field boundary maps.



Field polygons need to be simplified and made more regular to remove unnecessary vertices (e.g. stair steps along pixel edges), which can both improve map quality and reduce storage costs. A key limitation of models trained on non-VHR imagery is their inability to under-segment fields, leading to fields conjoined by narrow necks or overly complex, irregular shapes. Such shapes can be identified using geometric shape measures, as seen in the figure at left, where a model that uses field area and fractal dimension is used to identify undersegmented fields (red polygons).

In this focus area, we will identify and implement shape-based methods that can be used to both assess and improve the quality of FTW field boundary maps. We will do so by building and improving on

[approaches currently used](#) in our Mapping Africa workflow and downstream map-based assessments (Xiong et al, forthcoming).

Shape-based information and approaches for assessing and improving already generated maps will be further developed. Measures such as fractal dimension and compactness will be evaluated at larger scales to assess their reliability for identifying segmentation errors. Approaches for separating fields that are under-segmented (e.g. narrow-neck splitting) will be further developed.

The selected set of shape metrics will also be evaluated for their ability to assess relative performance improvements in field mapping models. These can be integrated as additional performance metrics in the model evaluation code, and also for assessing map quality over unlabelled regions. For example, a model that is more effective at separating adjacent fields should return a histogram of compactness values that are shifted further to the right than that of a less effective model. The shape metrics that indicate segmentation quality will likely vary between maps based on Sentinel-2 (FTW standard) and Planet (FTW Planet), thus we will also examine how shape metric-based map quality indicators vary by input dataset. Additionally, a lightweight model (e.g. Random Forests) based on shape metrics will be evaluated for efficacy in removing false positive fields.

Shape metrics may also prove useful within the model training process, by integrating them into the model's loss function. We have started exploring how to add a shape-based loss term to Tversky Focal Loss, and will continue to research this possibility under this project. This will include identifying existing approaches in the literature, such as from medical image segmentation, as well as formulating novel loss terms based on compactness and similar shape metrics.

Clark will be responsible for:

- Further research, testing, and selection of recommended shape metric suite for map quality assessment and reporting;
- Developing and testing polygon simplification, regularization, and splitting routines that can be integrated into the FTW ecosystem;
- Developing and testing shape metric-based models for flagging and removing false positives;
- Developing, testing, and potentially integrating shape-based loss terms with existing area-based loss functions;
- Demonstrating the utility of shape-based metrics in assessing model performance gains, post-inference map improvements, and for providing information on map quality and usability, through papers or other media.

Anticipated timelines

Efforts on the key focus areas and aspects of this project will overlap to a certain extent. Assuming an early to mid-summer start on labeling, the following time frames are anticipated:

- Focus Area 1: This will be ongoing and dependent on ASU's time frame for label collection, but the assumed start is July, 2026, with ongoing involvement through July, 2027;
- Focus Area 2: This tasks will commence in July, 2026 and conclude by July, 2027;
- Reporting and wrap-up of funded effort is anticipated to be completed by August, 2027. Beyond that timepoint, we expect to continue participating in ongoing FTW research meetings, through revision of submitted papers, and other related activities.